CHAPTER 2: Hardware & Software

Question 1: Summary of the Chip Designing & Manufacturing Video (Branch Education)

The video details the astonishing complexity of fabricating modern microchips (CPUs) housing billions of transistors:

Scale & Architecture: A modern CPU packs dozens of billions of transistors (e.g., 26 billion) into a silicon die the size of an ant. The architecture spans roughly 17 to 80 layers of microscopic copper wires (interconnects) stacked above nanoscopic FinFET transistors.

The Cleanroom Fab Environment: Silicon wafers (300 mm) travel in sealed front-opening pods (FOUPs) across an automated overhead track inside massive cleanrooms. Over a 3-month cycle, a single wafer undergoes roughly 940 to 1,000 steps.

Core Fabrication Cycle: The manufacturing is an iterative loop of ~~six~~ six primary tool operations:

Photolithography: Spin-coating photoresist and shining extreme UV light through expensive optical photomasks (stencils) to print nanometer-scale circuitry patterns.

Etching: Plasma or acid carving away exposed areas.

Deposition: Chemical or physical vapor deposition adding layers of copper, silicon dioxide insulators, or barrier metals.

CMP (Chemical Mechanical Planarization): Grinding the wafer flat with abrasive chemical slurry before the next layer begins.

Ion Implantation: ~~Blasting~~ Blasting dopants (boron, phosphorus) deep into silicon crystal lattices to form semiconductor junctions.

Metrology & Cleaning: Ultra-pure water rinsing and electron-microscope scanning to catch defects.

Testing & Binning: Completed wafers are sliced into dies, tested, and categoriped ("binned") into product tiers (i3, i5, i7, i9) depending on which functional cores survived manufacturing flaws, before final heat-spreader ~~packaging~~ packaging.

Question 2: Why did RAM get expensive in recent years?

RAM prices have seen dramatic spikes driven by structural supply-demand imbalances:

The AI & Data ~~Center~~ Center Boom: The rapid surge in LLMs and deep learning created overwhelming demand for high-bandwidth memory (HBM3e/HBM4) and high-density server

DDR5. DRAM manufacturers (Micron, SK Hynix, Samsung) shifted significant fab wafer allocation away from consumer DDR4/DDR5 to satisfy hyper-profitable enterprise AI orders.

Oligopolistic Supply: Over 90% of global DRAM production is controlled by just three manufacturers. When demand ~~fluctuates~~ fluctuates or capital expenditure is trimmed to protect margins, supply remains tightly constrained.

Extreme Fabrication Lead Times: Modern DRAM requires advanced EUV photolithography and cleanrooms costing tens of billions of ~~dollars~~ dollars. Increasing wafer capacity takes 2 to 3 years, preventing rapid supply adjustments during sudden demand surges.

Historical Supply Shocks: Past natural events (earthquakes in Taiwan), chemical contamination incidents, and geopolitical trade sanctions caused temporary manufacturing bottlenecks that rippled across global memory pricing.